The Reuleaux Platonic Solids

Theorem: with a ball as wide as the solid on every corner, the width along a face axis is exactly $2\sqrt{(3/4 + r^2)} - 2r$ and the smallest width is the diameter, so dual pairs agree exactly

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Version 1 · 21 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, checks and figure: https://github.com/DICTOR-Master/kaleidohedra/tree/0d42567/studies/2026-10-21-reuleaux-platonic (research branch, commit 0d42567; also attached to this record)
The construction (a ball on every corner of a Platonic solid, radius = the solid's diameter) and the names by look (Reuleaux octahedron, cube, icosahedron and dodecahedron) are DICTO's idea; computation, proofs and drafting with Claude (Anthropic).

Abstract. Main theorem. Put a ball on every corner of a Platonic solid of diameter 1, each of radius 1, and let B be what all the balls share (DICTO's idea). Then the smallest width of B is exactly 1, reached along any line through two corners at distance 1; and for the cube, octahedron, dodecahedron and icosahedron, with circumradius ½ and inradius r, the width along each axis through a face centre is exactly $2\sqrt{(3/4 + r^2)} - 2r$. Corollary: dual solids have the same ratio r/R, so dual pairs have exactly the same face-axis width, $(\sqrt{30} - \sqrt{3})/3 \approx 1.248392$ for the balls on the cube and on the octahedron and $(\sqrt{(50 + 2\sqrt{5})} - \sqrt{(5 + 2\sqrt{5})})/\sqrt{15} \approx 1.110989$ for the balls on the dodecahedron and on the icosahedron. The tetrahedron is the classical case, the Reuleaux tetrahedron, where the formula does not apply (its diameter is its edge, not twice its circumradius); its edge-midpoint axis gives $\sqrt{3} - \sqrt{2}/2$ by the same argument, and our volume agrees with the known closed form to 1.1×10^{-8}. The four new bodies are not Reuleaux polyhedra in Sallee's sense. Conjecture 1, with numerical evidence from 3000 directions and local refinement: these axes give the largest width. DICTO names the bodies by their look, the Reuleaux octahedron (balls on the cube), cube (on the octahedron), icosahedron (on the dodecahedron) and dodecahedron (on the icosahedron); each has the face pattern of the dual solid, and the fewest face colours are 4, 2, 3, 3 and 4.

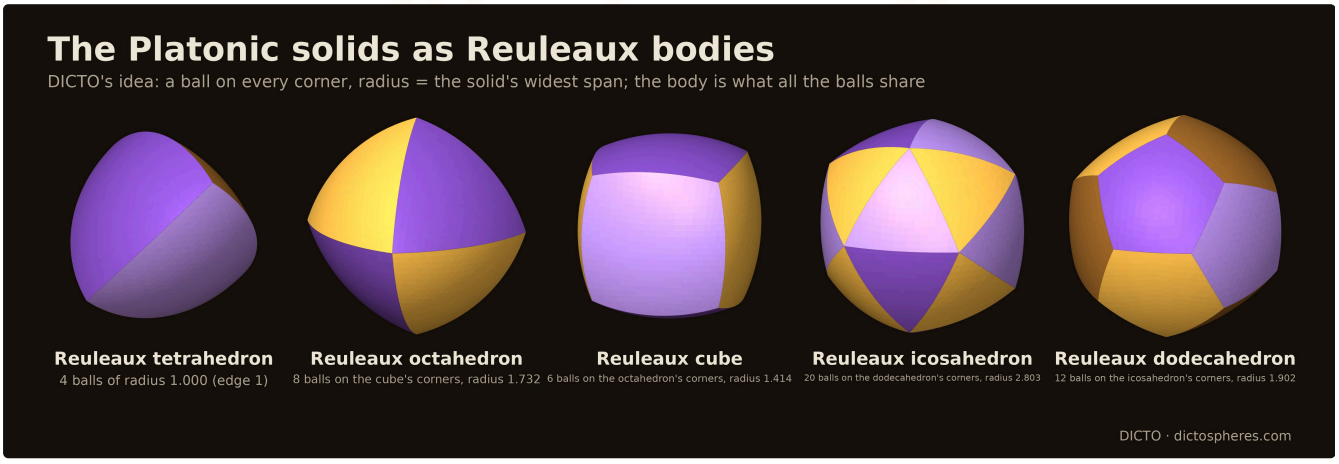


Figure 1. The five bodies at solid edge 1, each with its faces coloured with the fewest colours that keep neighbours apart (gold, purple, lilac, bronze). Under each: the number of balls and their radius, which is the solid's diameter. Meshes from reuleaux_platonic.py.

1. The theorem

For a Platonic solid S with corner set X and diameter D (its widest span), DICTO's body is B = $\cap_{v \in X}$ B(v, D), the common part of the closed balls of radius D centred on the corners; in the language of [1, 2] it is a ball polyhedron. For the tetrahedron D is the edge and B is the Reuleaux tetrahedron [2, 3]. For the cube, octahedron, dodecahedron and icosahedron D is twice the circumradius R: the widest spans join opposite corners. All lengths are at D = 1. The width of B in a direction u is the distance between the two planes at right angles to u that touch B.

> **Main theorem.** Let S be a Platonic solid of diameter 1 and B its body. (i) Every width of B is at least 1, and the width along the line through any two corners at distance 1 is exactly 1. (ii) If S is a cube, octahedron, dodecahedron or icosahedron, with circumradius ½ and inradius r, then along the axis through its centre and a face centre the width of B is exactly $2\sqrt{(3/4 + r^2)} - 2r$.

2. Proof

(i) By Pál's theorem, as cited in [2], the corner set X (of diameter 1) lies in a body K of constant width 1. K has diameter 1 and contains X, so every point of K is within 1 of every corner: K ⊂ B, and every width of B is at least 1. Now let v, w be corners with $|v - w| = 1$ and $u = v - w$. B lies in the ball of radius 1 about w, so $u \cdot x \le u \cdot w + 1 = u \cdot v$ on B, and v lies in B (no two corners are more than 1 apart); so the largest value of u·x on B is u·v, and in the same way the smallest is u·w. The width is $u \cdot v - u \cdot w = 1$.

(ii) Let u be the unit vector to the face centre and $h_v = u \cdot v$ the height of a corner v. The rotations of S about the axis (3-, 4- or 5-fold) map B to itself and fix u, so they map the set of points of B where u·x is largest to itself; that set is convex, and its centroid lies on the axis. So the largest value of u·x on B is reached at a point zu of the axis. Since $|v|^2 = ¼$, $|zu - v|^2 = ¼ + z^2 - 2z\,h_v$, and for $z > 0$ this is largest where h_v is smallest. The plane of the opposite face touches S, so every corner has $h_v \ge -r$, with equality exactly at the corners of that face. Hence zu lies in B exactly when $¼ + z^2 + 2zr \le 1$, that is $z \le \sqrt{(3/4 + r^2)} - r$. S and B are centrally symmetric, so the width is twice this. ∎

Verification. reuleaux_checks.py checks in exact arithmetic in $\mathbb{Q}(\sqrt{5})$ that the diameter of each of the four is 2R, that every corner has height at least −r along a face axis with equality exactly at the opposite face, the values of $(r/R)^2$, and the closed forms below; numerically, the face-axis widths equal the largest widths found by widths.py, and the smallest width found is 1 for all five.

3. Corollaries

Corollary 1 (dual pairs). The width in (ii) depends only on r = ½ × (r/R), and dual solids have the same r/R; the exact checks find $(r/R)^2 = 1/3$ for the cube and the octahedron and $(5 + 2\sqrt{5})/15$ for the dodecahedron and the icosahedron. So the bodies on a dual pair have exactly the same face-axis width:

- cube, octahedron: $r^2 = 1/12$, $2\sqrt{(3/4 + r^2)} = \sqrt{30}/3$ and $2r = \sqrt{3}/3$, so **$w = (\sqrt{30} - \sqrt{3})/3 \approx 1.248392$**;

- dodecahedron, icosahedron: $r^2 = (5 + 2\sqrt{5})/60$, $(2\sqrt{(3/4 + r^2)})^2 = (50 + 2\sqrt{5})/15$ and $(2r)^2 = (5 + 2\sqrt{5})/15$, so $\mathbf{w = (\sqrt{(50 + 2\sqrt{5})} - \sqrt{(5 + 2\sqrt{5})})/\sqrt{15} \approx 1.110989}$.

Corollary 2 (the tetrahedron, the classical case). Part (ii) does not apply to the tetrahedron: its diameter is its edge, not twice its circumradius, and it is not centrally symmetric (opposite a face lies a corner, not a face), so the width along a face axis is not twice the height reached on one side. On the tetrahedron B is the classical Reuleaux tetrahedron [2, 3]. Along the axis through the midpoints of two opposite edges the argument of (ii) does work (a quarter turn followed by a reflection maps the tetrahedron to itself and reverses the axis): the opposite edges are $1/\sqrt{2}$ apart and every corner is ½ from the axis, so the width there is $2\sqrt{(3/4)} - 1/\sqrt{2} = \mathbf{\sqrt{3} - \sqrt{2}/2 \approx 1.024944}$ (checked exactly).

Corollary 3 (not Reuleaux polyhedra). Hynd [3] and Montejano and Roldán-Pensado [2] call B(X) a Reuleaux polyhedron when X has the largest possible number of diametric pairs, $2m - 2$ for m points (the Vázsonyi bound, proved by Grünbaum, Heppes and Straszewicz [3]), or, after Sallee, when the vertices of B are exactly the points of X [2]. The tetrahedron has $6 = 2 \times 4 - 2$. The other four do not qualify (exact checks): their diametric pairs are only the opposite corners, 4, 3, 10 and 6, against 14, 10, 38 and 22; and each corner lies on the sphere of exactly one other ball (the opposite corner's), so it is not a vertex of B, where at least three spheres meet [2]. They are ball polyhedra, not Reuleaux polyhedra, and the smoothing of edge pairs that turns a Reuleaux polyhedron into a body of constant width [2, 3] does not apply to them as it stands.

4. Conjectures

> **Conjecture 1 (the largest width).** For the cube, octahedron, dodecahedron and icosahedron the largest width of B is the face-axis width of the main theorem; for the tetrahedron it is $\sqrt{3} - \sqrt{2}/2$, along the edge-midpoint axis.

Evidence (numerical). widths.py computes the width without a mesh, from the exact candidate points of the support function (ball caps, the circles where two spheres meet, the points where three meet), over 3000 random directions with local refinement from the six best. The largest widths found are 1.248391589161 for the balls on the cube and on the octahedron, 1.110988655456 for the balls on the dodecahedron and on the icosahedron, and 1.024944026382 for the tetrahedron: the values of the main theorem and of Corollary 2 to 12 digits. If Conjecture 1 holds, the width ranges of each dual pair, and so their spreads (largest/smallest), are equal, since the smallest width is 1 by (i).

5. Names by look, faces and colours

DICTO named the bodies by their look: balls on the cube's corners make the **Reuleaux octahedron**, on the octahedron's the **Reuleaux cube**, on the dodecahedron's the **Reuleaux icosahedron** and on the icosahedron's the **Reuleaux dodecahedron**. The look is explained by the faces. The surface of B is made of one spherical face from each ball (as for Reuleaux polyhedra [3]), lying across the body from its centre. The checks sample, for every pair of balls, the circle where their spheres meet, and find (numerically) that two faces meet along an arc exactly when their centres are joined by an edge of S. So the faces of B correspond to the corners of S and meet as those corners are joined, which is how the faces of the dual solid meet: the balls on the cube's 8 corners give 8 faces meeting like the octahedron's, and so on.

Colouring the faces so that no two neighbours share a colour is therefore colouring the corners of S along its edges. The fewest colours, found by exhaustive search, are 4 for the Reuleaux tetrahedron (any two of its 4 faces meet), 2 for the Reuleaux octahedron (the cube's corners split into two sets with no edge inside either), 3 for the Reuleaux cube, 3 for the Reuleaux icosahedron and 4 for the Reuleaux dodecahedron, as in Figure 1: the numbers DICTO read from the look, 4, 2, 3, 3, 4. The volumes (numerical, Table A1) include the tetrahedron check: Harbourne's formula $8\pi/3 + \sqrt{2}/4 - (27/4)\arccos(1/3)$, as quoted by Hynd [3], and our quadrature 0.4221577 agree to 1.1×10^{-8}.

6. Prior art, and the status of each result

- **Known, credited:** the Reuleaux tetrahedron and its volume (Harbourne, as quoted by Hynd [3]); ball polyhedra in general (Bezdek, Lángi, Naszódi and Papez [1]); Reuleaux polyhedra after Sallee [4], Meissner's solids and the Meissner polyhedra built from Reuleaux polyhedra (Montejano and Roldán-Pensado [2]); Pál's theorem (as cited in [2]); Hynd's perimeter and volume formulas for Reuleaux polyhedra, with a family of Reuleaux pyramids on regular n-gons for odd n [3]. The construction B(X) itself is the standard one of [1, 2, 3].
- **Proved in general (new here):** the main theorem for these bodies and Corollaries 1–3; the bodies on the cube, octahedron, dodecahedron and icosahedron and their names are DICTO's.
- **Proved for the cases built (exact checks):** the diametric pairs, the corners on one other sphere, the corner heights, $(r/R)^2$ and the closed forms.
- **Observed numerically:** Conjecture 1 (the largest widths); the faces meeting along arcs exactly along the solid's edges, and so the face patterns and fewest colours; the volumes.

The three library papers on these bodies [1, 2, 3] were searched in full for cube, octahedron, dodecahedron, icosahedron, Platonic and regular: the only regular solid they use is the regular tetrahedron or simplex (for the Reuleaux and Meissner tetrahedra, Hynd's pyramids and a lemma of [1]). Sallee's paper [4] is in the library as a link only and was not read. In the sources checked (the full text of the 81 library sources with a text layer listed in the appendix, searched on 21 October 2026; the library's 173 link-only entries, Sallee's among them, were not searched) we did not find the bodies on the other four Platonic solids or their widths.

Data and reproducibility

In studies/2026-10-21-reuleaux-platonic: reuleaux_checks.py (log reuleaux-checks.log, 45 PASS, no FAIL), in two parts. Part A is exact ($\mathbb{Q}(\sqrt{5})$ with Python fractions): the diameters, diametric pairs and the corners lying on one other sphere; the corner heights along a face axis; $(r/R)^2$ of each solid; the closed forms of the main theorem; and the tetrahedron's edge-midpoint axis. Part B is numerical (float64): the face-axis widths against widths.log, the volumes by radial Gauss–Legendre quadrature (N = 1000 and 2000 agree to within 6×10^{-7}), the Reuleaux tetrahedron's volume against Harbourne's formula, and the arcs (7200 points on each circle, margin 10^{-9}) with the fewest colours by backtracking. widths.py (widths.log): the width search. reuleaux_platonic.py and reuleaux_compose.py: the meshes (manifold3d), the colouring and Figure 1.

References

1. K. Bezdek, Z. Lángi, M. Naszódi, P. Papez, Ball-polyhedra (2007), arXiv:1110.4329.
2. L. Montejano, E. Roldán-Pensado, Meissner polyhedra (2017), arXiv:1608.06354.
3. R. Hynd, The perimeter and volume of a Reuleaux polyhedron (2023), arXiv:2310.08709.
4. G. T. Sallee, Reuleaux polytopes, Mathematika 17 (1970) 315–323, doi:10.1112/S0025579300002977 (link only; not read).

Appendix A. The five bodies

Body (DICTO's name)	Balls on	Faces	Arcs	Diametric pairs (2m − 2)	Fewest colours	Width range	Volume ≈
Reuleaux tetrahedron	tetrahedron	4	6	6 (6)	4	1 .. 1.024944	0.42216
Reuleaux octahedron	cube	8	12	4 (14)	2	1 .. 1.248392	0.64940
Reuleaux cube	octahedron	6	12	3 (10)	3	1 .. 1.248392	0.68764
Reuleaux icosahedron	dodecahedron	20	30	10 (38)	3	1 .. 1.110989	0.57188
Reuleaux dodecahedron	icosahedron	12	30	6 (22)	4	1 .. 1.110989	0.59679

Table A1. The five bodies at diameter 1. Faces: one per ball; arcs: pairs of faces meeting along an arc (numerical; equal to the solid's edges); diametric pairs: corner pairs at distance D, exact, with the bound 2m − 2 for m corners in brackets; fewest colours with no two neighbouring faces alike; width range: the minimum is proved (main theorem), the maximum is the largest width found numerically, proved to be attained (main theorem, Corollary 2) and conjectured to be the largest (Conjecture 1); volume by radial quadrature (numerical).

Appendix B. The reference library searched

The library's manifests list 267 entries, of which 94 have a local file; these give the 81 distinct sources with a text layer listed below and 13 page images. The full text of the 81 sources was searched on 21 October 2026 for: Reuleaux, Meissner, ball polyhedron, ball-polyhedra, constant width, intersection of balls, and, in the three papers that matched, cube, octahedron, dodecahedron, icosahedron, Platonic and regular. Only [1, 2, 3] matched the first terms. The 13 page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885) and the 173 link-only entries were not searched.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov (chapter 13, English translation)	1885	Elements of the Theory of Figures, chapter 13
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants

U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
J. H. Conway, C. Radin, L. Sadun	1999	On angles whose squared trigonometric functions are rational
R. V. Moody	2000	Model sets: a survey
I. Hafner, T. Zitko	2002	Rhombic polyhedra: from dissection of the cube to space filling with prolate rhombohedra and rhombic dodecahedra of the second kind
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2007-2015	Wolfram Demonstrations: 54-Faced Space-Filling Polyhedron (2007), Golden Octet Truss (2007), A Rhombic Dodecahedron Made of Rhombic Triacontahedra (2008), Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (2008), Cluster of 14 Rhombic Triacontahedra (2015)
K. Bezdek, Z. Lángi, M. Naszódi, P. Papez	2007	Ball-polyhedra
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity

L. Montejano, E. Roldán-Pensado	2017	Meissner polyhedra
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Hynd	2023	The perimeter and volume of a Reuleaux polyhedron
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)